

Paper 1 - Set 10 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Binomial expansion
2	B	Exponentials and logarithms
3	F	Angles, polygons, congruence and vectors
4	F	Sequences and series
5	A	Indices and surds
6	D	Integration
7	F	Polynomials, factor and remainder theorems
8	F	Pythagoras, mensuration and 3D
9	B	Sequences and series
10	A	Binomial expansion
11	B	Trigonometric functions and equations
12	B	Number, units and estimation
13	A	Graphs of functions
14	C	GCSE algebra
15	A	Exponentials and logarithms
16	E	Functions and transformations
17	B	Polynomials, factor and remainder theorems
18	C	Simultaneous equations and inequalities
19	C	Differentiation
20	F	Indices and surds

Worked solutions

Question 1 BIN-07 · Binomial expansion · spec MM2.4

Expand $(1 - 2x)^4$ and hence find the coefficient of x^2 in $(1 + 3x)(1 - 2x)^4$.

Answer: D

$(1 - 2x)^4 = 1 - 8x + 24x^2 - 32x^3 + 16x^4$. Coefficient of x^2 in the product: $1 \cdot 24 + 3 \cdot (-8) = 0$.

Question 2 LOG-18 · Exponentials and logarithms · spec MM5.2

If $\log_a b = 2$ and $\log_b c = 3$, find $\log_a c$.

Answer: B

$b = a^2$ and $c = b^3 = a^6$, so $\log_a c = 6$.

Question 3 GEO-01 · Angles, polygons, congruence and vectors · spec M5.2

The interior angle of a regular polygon is 156° . How many sides does it have?

Answer: F

Exterior angle 24° ; $360/24 = 15$ sides.

Question 4 SEQ-10 · Sequences and series · spec MM2.3

A geometric series has first term 2 and common ratio $r > 0$. The sum of the first three terms is 14. Find the sum of the first five terms.

Answer: F

$2 + 2r + 2r^2 = 14 \Rightarrow r^2 + r - 6 = 0 \Rightarrow r = 2$. $S_5 = 2(2^5 - 1) = 62$.

Question 5 IND-14 · Indices and surds · spec MM1.2

The number $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}}$ is equal to

Answer: A

$\sqrt{50} = 5\sqrt{2}$, $\sqrt{18} = 3\sqrt{2}$, $\sqrt{8} = 2\sqrt{2}$. So $\frac{8\sqrt{2}}{2\sqrt{2}} = 4$.

Question 6 INT-20 · Integration · spec MM7.2

Evaluate $\int_{-1}^2 |x| dx$.

Answer: D

Two triangles: $\frac{1}{2} \cdot 1 \cdot 1 + \frac{1}{2} \cdot 2 \cdot 2 = \frac{1}{2} + 2 = \frac{5}{2}$.

Question 7 POL-02 · Polynomials, factor and remainder theorems · spec MM1.6

$(x + 2)$ is a factor of $2x^3 + ax^2 - 5x - 6$. Find the value of a .

Answer: F

Factor Theorem: $f(-2) = -16 + 4a + 10 - 6 = 4a - 12 = 0$, so $a = 3$.

Question 8 MEN-14 · Pythagoras, mensuration and 3D · spec M5.16

A cylindrical tin of radius 4 cm and height 10 cm is full of water. The water is poured into an empty cylindrical jug of radius 5 cm. How deep is the water in the jug?

Answer: F

$$\pi \cdot 16 \cdot 10 = \pi \cdot 25 \cdot h \Rightarrow h = \frac{160}{25} = 6.4.$$

Question 9 SEQ-09 · Sequences and series · spec MM2.2

How many terms of the arithmetic series $3 + 7 + 11 + \dots$ are needed for the sum to first exceed 500?

Answer: B

$$S_n = \frac{n}{2}(6 + 4(n-1)) = n(2n+1). \quad n = 15: 465; \quad n = 16: 528 > 500. \text{ So } 16.$$

Question 10 BIN-05 · Binomial expansion · spec MM2.4

Evaluate $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$.

Answer: A

$$\binom{10}{3} = \binom{10}{7}, \text{ so the first two cancel; } \binom{9}{2} = 36.$$

Question 11 TRG-16 · Trigonometric functions and equations · spec MM4.6

Given that $\sin x + \cos x = 0$ and $0 \leq x < 2\pi$, find the sum of all possible values of x .

Answer: B

$$\tan x = -1: x = \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}. \text{ Sum} = \frac{10\pi}{4} = \frac{5\pi}{2}.$$

Question 12 NUM-16 · Number, units and estimation · spec M1.1

Water flows through a pipe at 1.2 litres per second. How long does it take to fill a tank of volume 3.6 m^3 ? ($1 \text{ m}^3 = 1000 \text{ litres}$.)

Answer: B

$$3600 \text{ litres at } 1.2 \text{ litres/s takes } 3000 \text{ s} = 50 \text{ minutes.}$$

Question 13 GRA-02 · Graphs of functions · spec MM8.7

The graphs of $y = x^2 - 2x$ and $y = k$ meet at exactly two points. Find the complete set of values of k .

Answer: A

$$y = (x-1)^2 - 1 \text{ has minimum } -1; \text{ a horizontal line above the minimum meets the parabola twice.}$$

Question 14 ALG-14 · GCSE algebra · spec M4.16

The sum of two numbers is 12 and the difference of their squares is 48. Find the larger number.

Answer: C

$$x^2 - y^2 = (x-y)(x+y) = 12(x-y) = 48 \Rightarrow x-y = 4. \text{ So } x = 8, y = 4.$$

Question 15 LOG-01 · Exponentials and logarithms · spec MM5.2

Find the value of $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$.

Answer: A

$$\log_2 \left(\frac{40}{5} \cdot \frac{1}{4} \right) = \log_2 2 = 1.$$

Question 16 FUN-15 · Functions and transformations · spec MM8.2

Which of the following is the equation of the graph obtained by stretching $y = x^3 - 3x$ parallel to the x -axis with scale factor $\frac{1}{2}$?

Answer: E

A stretch parallel to the x -axis with factor $\frac{1}{2}$ gives $y = f(2x) = (2x)^3 - 3(2x) = 8x^3 - 6x$.

Question 17 POL-12 · Polynomials, factor and remainder theorems · spec MM1.6

When $x^3 - 3x^2 + 2x + 7$ is divided by $x^2 + 1$, what is the remainder?

Answer: B

$x^3 - 3x^2 + 2x + 7 = (x^2 + 1)(x - 3) + (x + 10)$: expanding gives $x^3 - 3x^2 + x - 3 + x + 10$. Remainder $x + 10$.

Question 18 SIM-16 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $\frac{x-2}{x+1} \leq 0$.

Answer: C

A fraction is ≤ 0 when numerator and denominator have opposite signs (or the numerator is 0): $x - 2 \leq 0$ and $x + 1 > 0$ gives $-1 < x \leq 2$; $x - 2 \geq 0$ and $x + 1 < 0$ is impossible. $x = -1$ is excluded as the fraction is undefined.

Question 19 DIF-13 · Differentiation · spec MM6.1

$f(x)$ is a polynomial with $f'(x) = (x - 2)^2(x + 1)$. Which of the following is true?

Answer: C

f' changes sign from negative to positive at $x = -1$ (local minimum). At $x = 2$ the factor $(x - 2)^2$ does not change sign, so f' stays positive: a stationary point but not a turning point (a point of inflexion).

Question 20 IND-18 · Indices and surds · spec MM1.2

The expression $\frac{3}{7 - 2\sqrt{3}} - \frac{3}{7 + 2\sqrt{3}}$ simplifies to

Answer: F

Common denominator $(7 - 2\sqrt{3})(7 + 2\sqrt{3}) = 49 - 12 = 37$. Numerator: $3(7 + 2\sqrt{3}) - 3(7 - 2\sqrt{3}) = 12\sqrt{3}$.